Hydrogen UAV takeoff model

EMAE 355 · 2026

Engineering problem and work

Modeled the ground roll of a compressed-hydrogen fuel-cell UAV using force balance, lift, drag and rolling resistance. Explored changing vehicle mass as hydrogen is consumed.

Process

Physical assumptions → force model → numerical integration → sensitivity

Tools

MATLAB · Excel · Dynamics · Hydrogen

Scope and status

A modeling exercise, not a validated flight test

Portfolio case-study brief based on coursework and user-provided descriptions. This is NOT a copy of the original laboratory report. Diagrams on the website are explanatory illustrations, not photographs or original experimental plots.

Original source (may require Google Drive permission):
https://docs.google.com/spreadsheets/d/1sMx9B4HFBliU9oX2e4LINjH5vbJzH_dtenUp8BlWOrE/edit